

An Open-Source Modelling Framework for Forest Operational Planning

by

Rosalia Jaffray

B.S.F., The University of British Columbia, 2024

A RESEARCH PROPOSAL SUBMITTED IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
MASTER OF APPLIED SCIENCE

in

The Faculty of Graduate and Postdoctoral Studies

(Forestry)

THE UNIVERSITY OF BRITISH COLUMBIA

(Vancouver)

August 2026

© Rosalia Jaffray 2026

Abstract

The forest operational planning problem has long been recognized as a critical problem in forest resources management and its modelling. As a result, many models have been proposed as solutions to the specific sites and operational environments in question. These models have evolved with advances in data capability and computing power, but a consistent challenge remains; the given solutions have not been made reproduceable. A novel open-source modelling framework for forest operational planning was developed in Python. The main objectives of this modelling framework are to maintain open-source availability and accessibility with the goal of amplifying reproducibility and open innovation in forest operational planning. Using computational and quantitative measures, this framework will allow us to run and test scenarios that have not been tested before, allowing faster adaptation to the dynamic changes within our operational environments. For each of the presented framework and model, design considerations are carefully outlined and explained, allowing the approach to be used in a range of operational environments. Practical applications and implications of the open-source modelling framework are discussed.

Table of Contents

Abstract	iii
Table of Contents	iv
List of Tables	vi
List of Figures	vii
List of Abbreviations	viii
List of Symbols	ix
1 Introduction	1
1.1 Statement to Committee	1
1.2 Background	1
1.3 Problem Statement and Objectives	2
1.3.1 Research Questions and Objectives	2
1.4 Thesis Organization	3
1.4.1 Publications	3
2 Literature Review	6
2.1 Sustainable Forest Management Planning	6
2.2 The Operational Planning Problem	7
2.3 Accessibility in Operational Planning	8
2.4 Agility in Forest Operations	8
2.5 Small-scale Forest Operations in British Columbia	11
2.6 Gaps	12
3 Methods	13
3.1 Overview	13
3.2 Modeling Framework	13
3.2.1 Mathematical Model	13

Table of Contents

3.2.2	Parameters and Coefficients	16
3.2.3	Constraints	19
3.2.4	Objective Functions	21
3.2.5	Calibration and Validation	22
3.2.6	Computational Setup	23
3.2.7	Implementation	23
3.2.8	Deliverables within MASc Scope	23
4	Project Overview	24
	Bibliography	26

List of Tables

1.1	Chapter Breakdown	5
3.1	Core Components of the Model Formulation	15
3.2	Notations of the Model	17

List of Figures

List of Abbreviations

UBC	The University of British Columbia
BC	British Columbia
EEoP	Estimated End of Project
SFM	Sustainable Forest Management
SFO	Sustainable Forest Operations

List of Symbols

Σ summation

Chapter 1

Introduction

1.1 Statement to Committee

This proposal is a lab-level working document intended to accelerate committee alignment. MASc theses in our program do not formally require proposals; we nonetheless use them to codify objectives, risks, and deliverables early, acknowledging some ‘fuzzy parts’ that will be resolved iteratively. The student will proceed with high-confidence tasks (literature review; initial formulation; data assembly; skills ramp-up) while horizon length and certain design details progress through the stage-gates described herein.

1.2 Background

The overarching goal of the project is to develop a generalized, open-source operational planning model to be used in operational forestry practices. This open-source aim promotes accessibility, collaboration and innovation within the industry without a paywall barrier, supporting smaller-scale forestry operations across the province with a decision-support tool. This tool is not intended to replace or undermine the role of the operational forester, but rather support them in exploring the solution space with greater efficiency.

As different companies and countries all vary in their operational environment, there is a wealth of literature produced in forest operational planning methods but this knowledge generation is limited in its ability and accessibility for re-use. Henkelman (1978), for example, proposed a simulation model of timber harvesting operations in British Columbia (BC). Fast forward forty-six years and the foundational components of this model are still relevant and valuable. However, there have been dramatic changes in both data capture capabilities and computing power availability as well as approaches to forest management within this time that have evolved forest operations as a discipline (Heinimann, 2007). Henkelman (1978) proposed the

model as a means for examining change with the at-the-time current structure of forest operations in Gold River, BC. Change is a constant in forestry in British Columbia and as always, forestry in British Columbia needs to adapt. As do dynamic forest operations Bonita (1972). Some forest managers have larger areas and with that greater flexibility/agility/robustness in their forest management planning efforts, including in operational planning. Others, like small-scale forest operations or smaller area based tenures, may not have the same ability to pivot to other regions which incentivize them to make better management decisions. Making this argument will require more detail and is outside the feasibility of this research proposal draft.

1.3 Problem Statement and Objectives

The goal of this thesis is to advance the following objectives in the field of forest engineering:

- A generalized model for forest operational plan development
- Open-source access to operational planning tools

1.3.1 Research Questions and Objectives

The primary objective of this research project is to develop an open-source reusable modelling framework to address and improve accessibility and reproducibility of forest operations modelling, to better support future work of researchers and practitioners. This project proposes an open-source operational planning modelling framework for forest management tool development. The modelling framework is developed to answer this question:

- Is there a generalized formulation of the forest operational planning problem that can be adapted to different operational environments in British Columbia?

Secondary objectives and questions are to explore the applicability of this forest operational planning model framework as it anticipates and responds to future challenges, approaching this question:

- How can simulation modelling frameworks improve agility of forest operational planning in anticipation of future challenges?

Objective One: Conduct a systematic review of operations research-based tools for forest operational planning.

- Sub-question: How are operations research-based tools used for operational forest harvesting planning, and what are their strengths, limitations, and barriers to adoption?
- This review of current tools for forest operational planning provided a summary of strengths and limitations of the tools available, providing tangible informing of our design decisions of our framework to recognize these challenges and hopefully pose some useful options in the process.

Objective Two: Design and evaluate the applicability of a generalized open-source machine scheduling model for forest operational planning.

1.4 Thesis Organization

The thesis will be organized using the thesis-by-publication approach. The current presented structure is designed as an initial approach in a reflexive and iterative design style and is subject to change.

- Introduction
- Chapter One: A systematic review on forest operational planning models
- Chapter Two: An open-source generalized modelling framework for multi-integer linear programming model for forest operations scheduling
- Conclusion, limitations and future research directions

1.4.1 Publications

One of the aims of this project as an 18-credit thesis is to produce a minimum of one peer-reviewed journal article. Using the thesis-by-article approach, there are two planned articles from this project to be published. Key information for each chapter is presented in Table 1.1. Foss et al. (1998) identifies five groups of tasks/activities that comprise the modelling process with a supporting sixth group integrated throughout (that being documentation). The five tasks are: problem understanding, model representation, implementation and verification, initialization and debugging, and validation. The two journal articles and chapters to be presented in this thesis are organized using these tasks. Those interviewed by Foss et al. (1998) supported the idea that while experienced modelling is often unstructured (has

varying degrees of structure), there is a claim to be made that reminders of structure do not hinder the efficiency of such processes.

Chapter One

The planned title of this publication is "An open-source generalized modelling framework for multi-integer linear programming model for forest operations scheduling". This chapter will focus on the development and formulation of a generalized engine for exploring the forest operational planning solution space. This modelling framework can produce at greater efficiency a range of possibilities that forest managers can choose from, connecting technological development to the needs of the operational planning forester.

Chapter Two

The planned title of this publication is to be determined (see Thesis Organization for an attempt). This chapter applies and tests the validity of the modelling framework presented in Chapter One in a range of operational environments within British Columbia. Applying the framework to real-life case studies will allow us to generate multiple data points to map the trade-offs that exist within the solution space. Case studies will focus on small-scale forest operational environments. Discussion will highlight areas of opportunity to use these mapped trade-offs to communicate forest management decisions out to the public as well as add to the discussion of what 'good' operational plan options look like from an OR lens.

1.4. Thesis Organization

Table 1.1: Chapter Breakdown

Component	Chapter One	Chapter Two
Objective	Present open-source formulation of operational planning problem.	Test application of model on range of BC operational environments.
Relevance	Develop an accessible decision support system for operational foresters.	Validate model output in real-life case studies.
Data	VRI and publicly available data.	Case study inventory data.
Methods	Creation of MiLP modelling framework. Analysis (SWOT). Documentation of workflow and best practices.	Validation and analysis of outputs.
Target Journal	Croatian Journal of Forest Engineering (10 000 words)	International Journal of Forest Engineering (8000 words)
EEoP	To be discussed with KC.	To be discussed with KC.

Chapter 2

Literature Review

This research project focuses on the application of operations research (OR) methods within operational forest management planning.

2.1 Sustainable Forest Management Planning

Forest management planning is often described using hierarchical planning levels across different spatiotemporal scales: strategic, tactical and operational (Weintraub and Cholak, 1991). Strategic plans focus on long-term management of objectives across the landscape; in comparison, tactical plans have increased specificity over a shorter period and often include spatially linked information (Belavenutti et al., 2019). Tactical plans identify the general when and where questions associated with planning (Bettinger and Chung, 2004). It is operational plans, however, that focus on the specific implementation of these plans. Operational-level plans are focused on the spatially explicit allocation of resources to achieve the goals outlined in the strategic and tactical plans (Church, 2007). Operational-level decisions in forestry can span planning horizons of several months and can include hourly, daily, or weekly time-steps (Epstein et al., 2007). For greater detail in the comparison and key components of the different hierarchical planning levels, see (Ulvdal et al., 2023).

Across these multiple planning levels, there is a documented transfer of focus from timber-oriented planning to more holistic planning that includes a range of social, economic and environmental objectives (Weintraub and Bare, 1996). Researchers and practitioners alike have been vocal in their calls for change in sustainable forest management, initiating paradigm shifts for proactive management across multiple disciplines including wildfire coexistence (Daniels et al., 2024), hydrology and watershed management (Green and Alila, 2012), climate change (Schoene and Bernier, 2012) and social license (Moffat et al., 2016). These shifting approaches to forest management

have been documented in Polinko and Coupland (2021) bibliometric analysis of paradigm shifts in forestry. In the operational environment where forest operations can be used to aid in managing for these disturbances, timing constraints would be a critical factor just as they do in operational timelines (Bonita, 1972). In recent years, studies have focused in on the value of more holistic planning that incorporates uncertainty like Atkins et al. (2022) and Messier et al. (2015), adaptive forest operations (Tampekis et al., 2024) and capacity building and policy integration to link the various forest management planning levels more synergistically like Bogle and van Kooten (2015) and Gautam et al. (2017). Within this broader context, there has been growing focus on forest operations as all of these factors contribute to the environment where these operations are conducted. The ‘operational environment’ concept distilled in Röser (2012) further emphasizes the definition of forest operations itself as a product of social, environmental and economic factors. The evolving recognition of different values in forestry and the more holistic planning approaches in sustainable forest management and operations are echoed in the models used to inform these management decisions. Shifting trends in objective functions, model types and the embrace of multi-criteria decision analysis tools illustrates the evolution of sustainable forest management in both research and practice and their value in assessing viability of forest management planning efforts (Labarre et al., 2025).

2.2 The Operational Planning Problem

The aims of this section of the literature review are to describe the current state of literature focused on the operational planning problem, including:

- Definition of the operational planning problem
- Discussion of historical models in forest operational planning and the shifting modelling trends alongside evolving definitions of SFM and SFO
- Table showing shifting trends in modelling approaches and objective functions used in forest operational planning models (as related to the objective functions provided in the methodology section)

2.3 Accessibility in Operational Planning

Within forest operational planning specifically, the use of open-source modelling has been limited. Part of this challenge comes from difficulty in creating generalized models when there is a high level of site specificity and detail required in developing operational forest management plans (Rönnqvist et al., 2015). Approaches to forest operations, specifically, are dependent on a wide range of factors including site conditions, terrain and machine availability (Lahrsen et al., 2022). The operational constraints or thresholds of each machine define the environment that the machines can be used in; these constraints can be employed in decision support models to identify what specific harvesting systems can be used in a specific stand ((Visser et al., 2004);(Kühmaier and Stampfer, 2010);(Becker et al., 2018)). Bell and Keefe (2014) developed an operational costing model for forest operations; however, no open-source model has been built for operational scheduling and routing. While there is a long history and wealth of models developed to tackle this problem (See Table), the vast majority are developed with proprietary information and solved using black-box solvers on a discrete, custom and contract basis. This limited transparency reduces impact beyond individual corporations or management groups. In addition, many studies were conducted at times where the computational feasibility of the problems limited their application and scope (Rytwinski and Crowe, 2010). Oborn (1988) noted several outstanding issues that limited the effectiveness of forest operational planning models, including structural and interface limitations. Nearly three decades later, these issues continue to be relevant although there have been notable reductions in structural limitations with improvements to computing power availability and accessibility of certain modelling approaches (Martin et al., 2016). Interface limitations also affect the scalability (Paradis et al., 2018) and robustness (Myers, 2004) of forest operational planning models given that generally these are highly specific models that would require reworking if parameters were to change.

2.4 Agility in Forest Operations

In the varying spatio-temporal scales of forest management planning, there are also differences in actors' needs for agility on the landbase.

The concept of 'agility' in forest operational planning has been (Lebel et al., 2013) (Shahi and Pulkki, 2015) (Vergara et al., 2018)

2.4. Agility in Forest Operations

The importance of this data is recognized (Frayret and Perrier, 2016) but this actual ability to use this data is behind. The 'best' currently available data for forest management decisions is widely still the Vegetation Resource Inventory (VRI) that on landscape levels has higher accuracy but site for site may not be the level needed to conduct work.

Simulation tools can be an effective means in exploring agility in forest operational planning (Frayret and Perrier, 2016).

Foresters can be employed by a range of companies, from self-employment as a consultant to a small company or specialized contractor to working for a major licensee on a large tenure. Each of these types of forest management companies face different challenges in mathematical modelling skills development. For large forest management companies and major licensees, they have higher capital that can allow them to fund research projects and invest in more advanced modeling software. Some companies go as far as having their own specialized teams and expert support in-house to tackle complex modelling questions that other employees might have. These larger groups generally can host more training and professional development opportunities to grow other employees' skills in this area. However, these companies can also face difficulties like segregation of skills where their specialized teams handle the problems of higher complexity meaning that the more operational foresters do not need to have the expertise to fully understand the models used for planning. It also may mean that the operational foresters who know their jobs the best are being left out of some operational planning decisions that could greatly benefit from their input. Greater overhead and corporate limitations may reduce the speed at which new technologies can be adopted (i.e. more strict approach to best practices or protocols). Small-to-medium forest management companies often employ foresters who take on multiple roles within their job. This wearing of multiple hats means that they engage more directly with the decision-making and modelling rather than relying on a dedicated analytics team. Less corporate overhead also means that these companies may be more flexible in trying out new products or techniques. However, they might not have the financial capital to support expensive proprietary modelling tools or training for these tools, which may reduce the sophistication of modelling efforts as a function of time and cost limitations. Similar to independent contractors, cost and time limitations can hinder the adoption of new modelling approaches. Depending on the contractor, there may be little or high levels of hands-on experience with modelling but if so, they would need to be self-sufficient. For independent consultants, the time constraint to learning new models and tools is high given that

hours spent doing this are not spent doing other things that could make them more money. In both cases, the more custom approaches to modelling are underutilized because of these limitations. There is a greater reliance on heuristic-based block selection approaches (not talking about the modelling type but the bush skills, field judgement approach). The reliance on past experience and in-the-field scheduling for these company sizes dominate decision-making yet this decision-making is done with smaller margins and generally smaller land bases than the major licensee. Their decision-making often requires greater agility. From the perspective of a large-scale licensee, the involvement of the operational forester in model development and model explanation is critical in getting the operational forester to understand and trust the model outputs and insights. Cross-training between different skillsets rather than siloeing model development away from the system it is modelling could bridge the knowledge gap and improve operational efficiency. For small-to-medium companies and independent contractors, both groups would benefit from affordable access to modelling tools and their necessary training. Open-source forestry modelling tools provide an opportunity to reduce reliance on expensive, proprietary tools and encourage collaboration across academia, industry and government as well as across scales of forest management companies. In the context of sustainable forest management (what all of these companies and groups are hopefully working towards), confidence in decision-making is essential. The complexity of forest land management and the balance of SEE requires forest managers to navigate a dynamic operational environment with much uncertainty. By developing an accessible, customizable and scalable open-source modelling tool for decision support, operational foresters can enhance their ability to run and test counterfactual scenarios, allowing for data-driven, evidence-based decision-making that aligns with sustainability goals. They can make the same great decisions that they were making before but with greater exploration of the solution space. This allows them to explore and test new scenarios beyond their experience. All foresters operate within a restricted resource framework. For some groups, there are more constraints that concentrate the need for higher levels of confidence in their decision-making. One of these groups is small-to-medium forest management groups like community forests. With fewer hectares on their tenure compared to major licensees, there is less ability to absorb miscalculations. Their ability to implement sustainable forest management practices is not diluted across large holdings but is more concentrated. As explored above, these groups often lack access to high-cost proprietary tools. Confidence in forest management decisions stems from the ability to anticipate, evaluate and mitigate

risks. The ability to simulate different forest management alternatives, assess trade-offs and outcomes ensure that operational decisions are made off of the best available data and modelled projections, while also being supported by the heuristic assumptions and lived experience of the operational forester. The ability to simulate and test these alternatives can also provide tangible projections that can be used to communicate outward about the decision-making. Enhanced confidence can foster greater accountability and transparency, crucial components in building and maintaining social license.

2.5 Small-scale Forest Operations in British Columbia

Small-scale forestry operations across the province of British Columbia, like community forests and UBC’s own Research Forests, operate on smaller land-bases than industrial operations but continue to manage sustainably for a range of objectives across the land-base and the larger landscape. Community forestry practices plan and manage for a wide range of social, economic, cultural and environmental forest values and benefits, while prioritizing local decision-making across the various levels of forest management planning (Gunter, 2022). The increased focus on community-level management and smaller land-bases reduces the operational feasibility margins of these forests, which can in turn exacerbate the impact of ‘poor’ operational planning. Simply put, large industrial forest operations can go bid for better land somewhere else or manage some losses but the smaller, often area-based tenures have vested interest in reducing their economic risk, providing greater incentive for better forest management planning. This creates a strong incentive for more precise, well-planned forest management strategies that minimize risk and maximize long-term sustainability. However, while economic sustainability is a key driver, it is only one piece of the sustainable forest management and operations equation (Marchi et al., 2018). Effective operational planning must account for the ecological and social components of sustainability, as poor planning can lead to environmental degradation, loss of biodiversity and negative social impacts within the communities that depend on these forests (Tampekis et al., 2024). This context means that smaller mistakes in operational planning make it easier to sink these operations that have historically faced difficulty in implementation. Block for block, operational planning differences as compared to major licensees are comparable but on a full organizational-level, poor operational planning could be detrimental to the success of these operations (that are

doing great in approaching sustainable forestry at the ground-up level with community involvement, a range of objectives, etc.). The ability and vested interest in adapting to these changes are critical in the development of more community-focused sustainable forest management.

2.6 Gaps

The gap also needs to address why re-usability is important in mapping out the solution space, especially within smaller-scale context.

There is the need in this push for technology to continue to recognize the human-centered needs that these models are built for still (Bolis et al., 2025). This connection between work and worker links the role of ergonomics into the vying for industry 4.0 and 5.0, further integrating definitions like Marchi et al. (2018) and Röser (2012) who recognize the human-centered side of forest operations. Transparency and human involved development by those who use the model is a necessary part of building these products and their application.... into the argument of open-source and accessible tech for forest operational planning beyond that of the black-box solvers.

An open-source modelling framework presents a unique opportunity beyond cost savings; by providing an accessible framework developed for re-usability, organizations can track decision-making processes over time, learning what works and what doesn't in different scenarios (Dahlander et al., 2021). It increases ability to apply recourse in forest operational planning (Myers, 2004). There is also the ability to link in plausible emulations of operational planning into higher-level planning tools to anticipate operational level decisions at the strategic and tactical levels (Nelson et al., 1991). Linking planning levels to achieve sustainability is difficult in research, let alone in practice. There is also the promise of irreversible consequences of 'bad' operations that further demand the ability to explore operations and continue to action items amidst uncertainty. All of these big questions... but how do we get it done? There are also documented challenges of interoperability when linking components of the supply chain as well as mixing different spatial- temporal scales.

Chapter 3

Methods

3.1 Overview

3.2 Modeling Framework

3.2.1 Mathematical Model

An open-source modelling framework will be developed using Python (an open-source coding language). The open-source modelling framework will be developed according to best practices and methodology outlined in Lamprecht et al. (2020). Clear architecture for the open-source platform will be fleshed out by articulating currently available pieces, what is in the works and future developments that are possible. This initial articulation will take place before the framework and modelling is built within. Currently available pieces will be plugged in and a SWOT (Strength, Weakness, Opportunity, and Threat) analysis will be conducted for the open-source modelling framework.

Building off of the historical precedence of operations research methods that traditionally optimize time and profit in the forest operational planning problem, this open-source framework is designed with an emphasis on modularity. The separate but linked design structure allows the user to specify different objectives and their associated parameters without needing to redesign the whole database (Jones et al., 2001). This approach also is beneficial in the commercialization of the model, where proprietary custom-built add-on modules can be contracted out to allow more specific objectives and constraints to be modelled. The separation of different values is critical to make sure that you can adapt without need for major redesign if changing the objective function. For each objective function, it needs to match the coefficient units. This item needs to be actioned.

We will use a mixed-integer linear programming model (Kibzun and Rasskazova, 2023) to provide an optimal schedule for a given list of cutblocks (har-

3.2. Modeling Framework

vesting units) at the operational level (planning horizon of six weeks with shift-level time-steps). The given list of cutblocks will be from an identified list of planned blocks at the tactical level and their associated silvicultural prescriptions. Many studies have been developed to characterize blocks and provide insight into which harvest systems or equipment or choices should be used (Becker et al., 2018; Rytwinski and Crowe, 2010; Diaz-Balteiro and Romero, 2008) but this is beyond the scope of this study (could be a cool thing for integration or further research though). This will follow a rolling horizon where we run and then lock the first period and then add and re-run to create a simulation loop to show stability of the solution over time. This deterministic approach is done to empirically measure depth/length for re-planning stably. Proper notation will be denoted for all components. There will be an explanation of the objectives, parameters and constraints in paragraph form. Presentation of the mathematical formulation will be in the standard form followed within the FRESH Lab.

The core components of the model are as described below. We will then require a matrix to link stands with time periods and operations for decision variables (complete action a in stand i in period t). A separate matrix will also need to be designed for roads, linking decision variables to road development activities (conduct action x for road a in period t). Calculating the dollars per scheduled machine hour, we can calculate our opportunity cost. The equation is $SMH = PMH/\text{machine utilization}$ (Holzleitner et al., 2011). The combination of block, machine and prescription will yield a single productivity and inventory volume that can be used in the model.

Assumptions of the mathematical model are as follows:

- There are a set number of harvesting activities. The list of activities and associated machines are presented in a table.
- There is a virtual depot that helps with movement. It is like a virtual nap spot for the equipment to return to every day.
- Block is already designed for a specific system and therefore this is predetermined and not a decision to be made in the model
- Sequencing according to absolute area or volume limits depending on the system (Buncher/Skidder: 2000 m³ AND Harvester/Forwarder: 120 m³)

The model will be designed using precedent relationships (Arora, 2023), given that certain equipment cannot operate on a stem if other activities

3.2. Modeling Framework

Table 3.1: Core Components of the Model Formulation

Component	Required Information	Further Questions
Stands/Harvest Units	Spatial data, area. Stand type. Seral stage.	Inclusion of operability information (slope, etc).
Time Periods	Definition of horizon and time-steps.	Justification for different harvest systems.
Operational Activities	Description of forest mgmt actions that can be done in each period in each polygon.	Road build and maintenance included?
Roads	ID. Geometry. Length. Status. Volume.	Decision var linked to road dev activities.
Equipment (var productivity)	Site specific conditions, what Rx they can do	Cost, capacity, operational time, flexibility (Arora, 2023)

have not first been completed. In simple terms, you cannot buck a tree that hasn't been felled. To trigger the availability for one activity to start after another, working inventory can be used to model changes in volumes. If the productivity of primary transport is greater than the productivity of felling, then there would be a need to slow or limit the availability of primary transport to meet the requirements. Slowing this could look like an inventory buffer cap. In the event that there are piling up volumes of felling, then it would trigger greater availability of primary transport until maximum capacity.

If there is a small given machine set, this is a highly constrained combinatorial optimization problem. With a larger machine set, there is a much bigger challenge.

The objective of the model is to generate an optimal operational-level schedule (three-month planning horizon with shift-level time steps) for a set of cut blocks using available equipment and labour. The mathematical formulation

3.2. Modeling Framework

of the proposed model is as follows:

$$\min \text{ or } \max \sum_{t \in T} \sum_{m \in M} \sum_{b \in B} \sum_{p_r \in P_R} \sum_{p \in P} c_{tmbr_p} x_{tmbr_p} \quad (3.1)$$

s.t.

Machine Use Decision Variable

Let x_{t,m,b,r_p} be a binary decision variable, where:

$$x_{t,m,b,r_p} \in \{0,1\} \forall t \in T \quad \forall m \in M \quad \forall b \in B \quad \forall r_p \in R_P \forall p \in P$$

If machine m works on activity p in block b during time t , then 1, otherwise 0.

Machine Movement Decision Variable

Let $y_{m,i,j,t}$ be a binary decision variable, where:

$$y_{m,i,j,t} \in \{0,1\} \quad \forall m \in M \quad i,j \in B \quad t \in T$$

If machine m moves from block i to j at time t , then 1, otherwise 0.

Volume Decision Variable

Let $I_{b,p,t}$ be a binary decision variable, where:

$$I_{b,p,t} \geq 0$$

This denotes the available inventory volume for activity p at block b at time t .

Add scheduling arrival and finishing times variables to act as constraints
Inventory by productivity is duration if calculated with utilization and SMH
Add size of block as a parameter (ha)

3.2.2 Parameters and Coefficients

Parameters (fixed values) and coefficients (unit values) will depend on the core components. Below you can find further descriptions of the included parameters and coefficients.

- Adjacency and Spatial Linkages

3.2. Modeling Framework

Table 3.2: Notations of the Model

Notations	Description
<i>Sets</i>	
T	Set of time periods (shifts) $i \in N_T$
M	Set of machines $i \in N_M$
B	Set of blocks (from tactical list) $i \in N_B$
R	Set of prescriptions $i \in N_R$
P	Set of harvesting activities (felling, primary transport, processing) $i \in N_P$
$P_r \subset R$	Set of activities required by prescription $r \in R$
W	Set of worker types $i \in N_W$
$\mathcal{N} \subset B \times B$	Set of block pairs (near)
$\mathcal{F} \subset B \times B$	Set of block pairs (far)
<i>Parameters</i>	
$\text{prod}_{m,p,b}$	Productivity rate of machine m for activity p in block b
$I_{b,p}$	Operable volume for prescription r in block b
$\text{vol}_{b,p}$	Total volume required for activity p to start in block b
$\text{op}_{m,(i,j)}$	Operating cost for machine m to perform activity p
$\text{mob}_{m,(i,j)}$	Mobilization cost of moving machine m from i to j
$\text{avail}_{m,t}$	Availability of machine m at time t
$\text{req}_{w,m,p}$	Required number of workers of type w to operate machine m for activity p
$\text{avail}_{w,t}$	Availability of worker w at time t
$a_{b,r}$	Combination of Silvicultural prescription assigned to block b (1,0) binary
<i>Variables</i>	
$x_{t,m,b,r_p,w}$	Machine assignment variable (binary)
$y_{m,i,j,t}$	Machine movement variable (binary)
$I_{b,p,t}$	Inventory volume variable (continuous)

- Constraints: definitions and parameters
- Objective Functions: definitions and coefficients
- Equipment/Workforce: time availability, equipment availability, workforce availability and assignment
- Productivity rate of machine m for activity p
- Mobilization cost of moving machine m from i to j

Blocks and Block Information

The list of blocks to be used in the model come from a tactical list of blocks and have assigned silvicultural prescriptions. These components are not decisions within the model but if you'd like, there are some models that can do that but not ours (Becker et al., 2018; Rytwinski and Crowe, 2010; Diaz-Balteiro and Romero, 2008). Part of the available block information will be gleaned from the available VRI data whereas other information like adjacency between blocks will be represented using the near/far description presented in Mobilization. Adjacency from a long-term planning perspective is outside the scope of this model as the tactical list of blocks within a set period of time would have already accounted for that constraint, using a strategic planning forest estate model.

Piece Size Calculation

Piece size is the size of individual trees or logs that are harvested and is recognized as a critical factor in machine productivity (Lahrsen et al., 2022). It can be calculated using available VRI attribute information. It is calculated by dividing stem density (stems/ha) by merchantable volume ($\text{m}^3 \text{ha}^{-1}$). Assuming that both stem density and merchantable volume are included as VRI attribute data columns, we propose to use these values to compute piece size for use in the productivity function. Where data gaps exist, we can supplement both stem density and merch volume estimates from VDYP and TIPSYS growth and yield simulation outputs.

Productivity function

Different productivity rates will be provided as coefficients of machine m for activity p in prescription r . Productivity is represented as the volume over productive machine hours. Utilization is represented as productive machine

hours over scheduled machine hours. The relationship of machines and productivity is mapped using productivity equations, that are based on block attributes. Depending on the category and type of machine, productivity may differ. Productivity equations are fed with a variety of factors including piece size, terrain characteristics (e.g. slope) among others. Machine utilization is set by the user. Factors that influence productivity are including but not limited to: ground slope and rugosity, risk of rutting or other soil damage issues, stem density, piece size, tree species, sorting and piling policy logic complexity, prescription type and intensity, machine type and model (as compiled from Lahrsen et al. (2022) and Holzleitner et al. (2011)). When designing the model, X and X were specified as factors to be included in the productivity equations given the prevalence and availability of this information across almost all blocks that would be harvested. Cruising summaries and basic terrain information is a fundamental input in block design and therefore, using this information in the productivity equations does not self-select or eliminate some blocks more than others from being included and assessed properly in the model. Different harvesting systems will have different factors that impact the overall productivity and in BC, there is a wide range of treatments and sites that can impact productivity. For the purpose of this analysis, we will focus on the following specific site and treatment types. Further studies may include treatments and sites such as X, Y, Z. Productivity is often represented as functions that can range from simple to more complex. The model will be designed using simple productivity functions that are made specific using the input data. Maintaining stable functions across the models allows for the needed specifications within each case instead of embedding the specifics within the model itself, reducing its ability to pivot and explore alternate solution spaces.

Description of real-world productivity characteristics

- General
- Specific to BC
- Site characteristics

3.2.3 Constraints

Mobilization

In the model, mobilization is included as a constraint. For simplification, two sets of blocks are presented: near (blocks are within walkable distance

3.2. Modeling Framework

(≤ 1 km) of each other (Arora, 2023)) and far (trucking is required, incurring an additional cost). The additional cost incurred is described as the parameter, mobilization cost of moving machine m from i to j . The distances between the blocks are identified as the shortest path distances using ArcGIS, using the available connected road network. When this distance is near, no mobilization cost. When far, cost incurred.

$$x_{m,j,k,t} \quad (3.2)$$

Precedence Relationships

Harvesting tasks need to be conducted in a specific order. For example, trees need to be felled before they can be skidded before they can be processed before they can be loaded. For two activities p and p' , if p must precede p' , then:

Multi-Task Machine Capability

Some machines are able to conduct multiple activities. However, they can only conduct one activity p at a given time t .

Machine Availability

Each machine can only be doing one thing at a time in one place. Need to include workers. Specify for each worker and machine.

$$\sum_{b \in B} \sum_{r_p \in R_P} x_{m,b,p,t,w} \leq 1 \quad \forall m \in M, \forall t \in T, \forall w \in W \quad (3.3)$$

Number of machines assigned to job x cannot exceed number of machines trained to job x .

$$\sum \quad (3.4)$$

For each time step, with summation on machine and worker How many machines available to do any type of job? For each worker and each time

Labour Availability

Each worker can only be doing one thing at a time in one place. For every shift, the total number of workers needed for all machines and activities cannot exceed how many workers of each type are actually available. Number

3.2. Modeling Framework

of workers assigned to job x cannot exceed number of workers trained to job x .

$$\sum_{m \in M} \sum_{b \in B} \sum_{p \in P} \text{req}_{w,m,p} \cdot x_{m,b,p,t} \leq \text{avail}_{w,t} \quad \forall w \in W, \forall t \in T \quad (3.5)$$

Total Volume Completed

All required work (volume) needs to be completed for each activity in each block.

$$\sum_{t \in T} \sum_{m \in M} \text{prod}_{m,k} \cdot x_{m,b,p,t} = \text{vol}_{b,p} \quad \forall b \in B, p \in P_{r_b} \quad (3.6)$$

Inability to Schedule Non-Prescribed Activities

The model is unable to schedule activities that are not explicitly identified in the set of activities allocated to the prescription assigned to the block.

$$x_{m,b,p,t} = 0 \quad \forall b \in B \quad \forall p \notin P_{r_b}, m \in M, t \in T \quad (3.7)$$

Mobilization Between Blocks

$$\sum_{j \in J} \sum_{m \in M} y_{m,i,j,t} \leq |M| \quad \forall t \in T, \quad \forall i \in I \quad (3.8)$$

3.2.4 Objective Functions

The model is generic. We will present multiple objectives (or flavours of modelling) to be tested and workable within the model to map out the needs of the user. The model will provide summations of scalar products of the decision variables by coefficients assigned to each objective function. Some possible objective functions include:

- **Minimization Objective Functions**

- Harvest Cost (e.g., \$/m³)
- Operational Time (e.g., h)
- Downtime for Machinery / Max Utilization (e.g., % or h)
- Soil Erosion or Sediment Yield (e.g., t ha⁻¹ or kg m⁻³)
- Environmental 'Damage' Badness (unitless or index score)
- Carbon Emissions from Operations (e.g., kg[C]/m³)

- Deviation from Even-Flow (e.g., %)

- **Maximization Objective Functions**

- Multi-Criteria Utility (unitless – composite score or index) (Diaz-Balteiro and Romero, 2008)
- Volume Harvested (e.g., m³)
- Workforce Utilization (e.g., % or worker-hours)
- Net Present Value (e.g., \$)

I plan to provide a table showing dominant citations for each of the objective functions that will be presented within the modelling framework and highlight the trends of these objective functions throughout time as the definition of SFO evolves. This information will be presented in my literature review and will inform the development of the framework. If we were to assess environmental 'badness' or damage, this could be some combination of machine/block information related to rutting, streams, residual damage that would create an environmental penalty cost.

3.2.5 Calibration and Validation

The current standard of forest operational planning is done by operational registered professional foresters; the current state is already at an impressive state of efficiency but opportunity remains for forest management to be more adaptive and less based on the past practices. The model will be calibrated with historical operational plans developed by operational registered professional foresters to confirm that the presented framework and simulations can effectively model the realistic system. From there, we can introduce components of this changing environment to see the differences in the counterfactual. The model will be validated using three operational environments in British Columbia: interior, Vancouver Island and coastal mainland. For each of these operational environments, the following components will be compiled: site characteristics and description, experimental design, and comparative visualization (Janová et al., 2024). The what-if scenarios will be compared against the baseline scenario for alternative scenario analysis.

This validation and scenario analysis will act as the basis for the foundation of mapping the trade-offs between different operational planning approaches.

3.2.6 Computational Setup

Computationally, [description of required resources]. For the optimization procedure, it is anticipated that at scale these models may exceed black-box solver capacity and meta-heuristic optimization methods may be needed. Depending on the outcomes, the computational intensity may be warranted or make it infeasible.

3.2.7 Implementation

For implementation and accessibility, consider the following:

- Versioning Control
- Naming Conventions
- Referencing
- Modular design
- Indexing/foreign keys
- Usable documentation according to FAIR Principles (Lamprecht et al., 2020)
- Presentation of Summary for Policymakers (SPM) at the end of the thesis
- Following similar format for community reports to generate usable, comprehensive information about the model (with decent, readable length)
- Glossary (see (Preston, 2024) for style preference)

3.2.8 Deliverables within MASc Scope

Deliverable is a research-grade prototype with reproducible cases and documented limits; not a full commercial tool. Clear path for future lab iterations; this student is not expected to deliver polished “pro-grade” UX/hardening.

Chapter 4

Project Overview

As an open-source solution, it democratizes access to advanced planning tools, fostering innovation and continuous improvement across the forestry sector (Dahlander et al., 2021). The operational harvest scheduling tool developed as my first paper will serve as a foundational component of the integrated framework developed by my team (FRESH Lab) to cover both harvesting and logistics operations, enabling accurate scenario modeling and analysis of the counterfactual without incurring implementation costs. This is nested within a broader appreciation that the computational approaches will allow us to explore what we haven't before using this proposed combination of computational and quantitative measures. This framework can be used to test 'what-if' scenarios of the counterfactual, allowing forest managers greater flexibility in assessing the impact of innovative ideas (e.g., changes to equipment, different scheduling, impact of fires) on the operational planning. These contributions (tools) can inform decision-making, allowing current operational planning strategies to be modelled and compared with the outputs of different management alternatives.

Developing this model lays the groundwork for complementary research with open-source trucking logistic models, permitting future research that can investigate the impact of harvesting decisions on transportation and vice versa. For example, how does the spatial distribution of harvesting impact transportation costing or route scheduling? Inversely, additional studies could explore the impact of different trucking and fleet configurations (e.g. percent electric, payload limitations) on optimal harvest scheduling design. This collective knowledge could lead to industry-wide advancements, allowing operational efficiencies and environmental sustainability to evolve in tandem across different forest management contexts with greater agility and lower barrier-to-entry. This project is an initial step towards framework development that can function within the complex dynamic system of forest operational planning.

The smaller the operator, the more exposed they are to statistically fuzzy inventory data that they use in operational planning processes. The ability to quickly explore feasible alternatives (recourse) is critical. Based on this argument, there are future areas for exploration:

- Cutting permits and regulatory approaches
- Different harvesting systems and integration with harvest system allocation
- Costing integration (Bell and Keefe, 2014)

Bibliography

- Arora, R. (2023). *Optimization of forest harvest scheduling at the operational level*. PhD thesis, University of British Columbia.
- Atkins, J. W., Stovall, A. E., and Silva, C. A. (2022). Open-source tools in r for forestry and forest ecology. *Forest Ecology and Management*, 503:119813.
- Becker, R. M., Keefe, R. F., Anderson, N. M., and Eitel, J. U. (2018). Use of lidar-derived landscape parameters to characterize alternative harvest system options in the inland northwest. *International Journal of Forest Engineering*, 29(3):179–191.
- Belavenutti, P., Romero, C., and Diaz-Balteiro, L. (2019). Integrating strategic and tactical forest-management models within a multicriteria context. *Forest Science*, 65(2):178–188.
- Bell, C. K. and Keefe, R. F. (2014). Fvs-opcost: A new forest operations cost simulator linked with fvs. In *Proc. of the 37th Council of Forest Engineering Conference annual meeting, Virginia Tech College of Natural Resources and Environment*.
- Bettinger, P. and Chung, W. (2004). The key literature of, and trends in, forest-level management planning in north america, 1950–2001. *International Forestry Review*, 6(1):40–50.
- Bogle, T. and van Kooten, G. C. (2015). Protecting timber supply on public land in response to catastrophic natural disturbance: a principal-agent problem. *Forest Science*, 61(1):83–92.
- Bolis, I., Marques, J. N., Cagno, E., and Morioka, S. N. (2025). Digital technologies, sustainability, and work: How can these themes be brought together to promote a human-centered future in industry 5.0 implementation? *Applied Ergonomics*, 125:104475.

- Bonita, M. L. (1972). *A simulation model for planning and control of forest harvesting operations*. PhD thesis, University of British Columbia.
- Church, R. L. (2007). Tactical-level forest management models. In *Handbook of operations research in natural resources*, pages 343–363. Springer.
- Dahlander, L., Gann, D. M., and Wallin, M. W. (2021). How open is innovation? a retrospective and ideas forward. *Research Policy*, 50(4):104218.
- Daniels, L. D., Dickson-Hoyle, S., Baron, J. N., Copes-Gerbitz, K., Flannigan, M. D., Castellanos-Acuna, D., Hoffman, K. M., Bourbonnais, M., Wilkinson, S. L., Roeser, D., et al. (2024). The 2023 wildfires in british columbia, canada: impacts, drivers, and transformations to coexist with wildfire. *Canadian Journal of Forest Research*, 55:1–18.
- Diaz-Balteiro, L. and Romero, C. (2008). Making forestry decisions with multiple criteria: A review and an assessment. *Forest ecology and management*, 255(8-9):3222–3241.
- Epstein, R., Karlsson, J., Rönqvist, M., and Weintraub, A. (2007). Harvest operational models in forestry. *Handbook of operations research in natural resources*, pages 365–377.
- Foss, B. A., Lohmann, B., and Marquardt, W. (1998). A field study of the industrial modeling process. *Journal of Process Control*, 8(5-6):325–338.
- Frayret, J.-M. and Perrier, N. (2016). Introduction to agility in the forest product value chain.
- Gautam, S., LeBel, L., and Beaudoin, D. (2017). A hierarchical planning system to assess the impact of operational-level flexibility on long-term wood supply. *Canadian Journal of Forest Research*, 47(4):424–432.
- Green, K. C. and Alila, Y. (2012). A paradigm shift in understanding and quantifying the effects of forest harvesting on floods in snow environments. *Water Resources Research*, 48(10).
- Gunter, J. (2022). Community forestry in british columbia, canada: History, successes, and challenges. In *Routledge Handbook of Community Forestry*, pages 319–337. Routledge.
- Heinimann, H. R. (2007). Forest operations engineering and management—the ways behind and ahead of a scientific discipline. *Croatian Journal of Forest Engineering: Journal for Theory and Application of Forestry Engineering*, 28(1):107–121.

- Henkelman, L. A. (1978). *GPSSV simulation model of timber harvesting operations*. PhD thesis, University of British Columbia.
- Holzleitner, F., Stampfer, K., and Visser, R. (2011). Utilization rates and cost factors in timber harvesting based on long-term machine data. *Croatian Journal of Forest Engineering: Journal for Theory and Application of Forestry Engineering*, 32(2):501–508.
- Janová, J., Bödeker, K., Bingham, L., Kindu, M., and Knoke, T. (2024). The role of validation in optimization models for forest management. *Annals of Forest Science*, 81(1):19.
- Jones, J., Keating, B., and Porter, C. (2001). Approaches to modular model development. *Agricultural Systems*, 70(2):421–443.
- Kibzun, A. I. and Rasskazova, V. A. (2023). Linear integer programming model as mathematical ware for an optimal flow production planning system at operational scheduling stage. *Automation and Remote Control*, 84(5):529–542.
- Kühmaier, M. and Stampfer, K. (2010). Development of a multi-attribute spatial decision support system in selecting timber harvesting systems. *Croatian Journal of Forest Engineering: Journal for Theory and Application of Forestry Engineering*, 31(2):75–88.
- Labarre, C., Domec, J.-C., Andrés-Domenech, P., Bödeker, K., Bingham, L., and Loustau, D. (2025). Improving forest decision-making through complex system representation: A viability theory perspective. *Forest Policy and Economics*, 170:103384.
- Lahrsen, S., Mologni, O., Magalhães, J., Grigolato, S., and Röser, D. (2022). Key factors influencing productivity of whole-tree ground-based felling equipment commonly used in the pacific northwest. *Canadian Journal of Forest Research*, 52(4):450–462.
- Lamprecht, A.-L., Garcia, L., Kuzak, M., Martinez, C., Arcila, R., Martin Del Pico, E., Dominguez Del Angel, V., Van De Sandt, S., Ison, J., Martinez, P. A., et al. (2020). Towards fair principles for research software. *Data Science*, 3(1):37–59.
- Lebel, L., Gautam, S., and Beaudoin, D. (2013). Agility capabilities in wood procurement systems: a literature synthesis. *International journal of forest engineering*, 24:216–232.

- Marchi, E., Chung, W., Visser, R., Abbas, D., Nordfjell, T., Mederski, P. S., McEwan, A., Brink, M., and Laschi, A. (2018). Sustainable forest operations (sfo): A new paradigm in a changing world and climate. *Science of the Total Environment*, 634:1385–1397.
- Martin, T., Hofman, J. M., Sharma, A., Anderson, A., and Watts, D. J. (2016). Exploring limits to prediction in complex social systems. In *Proceedings of the 25th international conference on world wide web*, pages 683–694.
- Messier, C., Puettmann, K., Chazdon, R., Andersson, K., Angers, V. A., Brotons, L., Filotas, E., Tittler, R., Parrott, L., and Levin, S. A. (2015). From management to stewardship: viewing forests as complex adaptive systems in an uncertain world. *Conservation Letters*, 8(5):368–377.
- Moffat, K., Lacey, J., Zhang, A., and Leipold, S. (2016). The social licence to operate: a critical review. *Forestry (London)*, 89(5):477–488.
- Myers, J. A. (2004). *Analysis of Robustness of Operational*. PhD thesis, University of New Brunswick.
- Nelson, J., Brodie, J. D., and Sessions, J. (1991). Integrating short-term, area-based logging plans with long-term harvest schedules. *Forest Science*, 37(1):101–122.
- Oborn, R. M. (1988). Towards a terramechanics framework in evaluating vehicle concepts for forest off-road locomotion.
- Paradis, G., Bouchard, M., LeBel, L., and D’Amours, S. (2018). A bi-level model formulation for the distributed wood supply planning problem. *Canadian Journal of Forest Research*, 48(2):160–171.
- Polinko, A. D. and Coupland, K. (2021). Paradigm shifts in forestry and forest research: a bibliometric analysis. *Canadian Journal of Forest Research*, 51(2):154–162.
- Preston, G. (2024). *Me7 tsertsrép re nekéct: me tsecéntem re tmicw-kt*. PhD thesis, University of British Columbia.
- Rönnqvist, M., D’Amours, S., Weintraub, A., Jofre, A., Gunn, E., Haight, R. G., Martell, D., Murray, A. T., and Romero, C. (2015). Operations research challenges in forestry: 33 open problems. *Annals of Operations Research*, 232:11–40.

- Röser, D. (2012). Operational efficiency of forest energy supply chains in different operational environments. *Dissertationes forestales*, 146:83.
- Rytwinski, A. and Crowe, K. A. (2010). A simulation-optimization model for selecting the location of fuel-breaks to minimize expected losses from forest fires. *Forest ecology and management*, 260(1):1–11.
- Schoene, D. H. and Bernier, P. Y. (2012). Adapting forestry and forests to climate change: A challenge to change the paradigm. *Forest Policy and Economics*, 24:12–19.
- Shahi, S. and Pulkki, R. (2015). simulation-based optimization approach to integrated inventory management of a sawlog supply chain with demand uncertainty. *Canadian journal of forest research*, 45(10):1313–1326.
- Tampekis, S., Kantartzis, A., Arabatzis, G., Sakellariou, S., Kolkos, G., and Malesios, C. (2024). Conceptualizing forest operations planning and management using principles of functional complex systems science to increase the forest’s ability to withstand climate change. *Land*, 13(2):217.
- Ulvdal, P., Öhman, K., Eriksson, L. O., Wästerlund, D. S., and Lämås, T. (2023). Handling uncertainties in forest information: the hierarchical forest planning process and its use of information at large forest companies. *Forestry*, 96(1):62–75.
- Vergara, F. P., Palma, C. D., and Nelson, J. (2018). Modeling lean and agile approaches: A western canadian forest company case study. *Forests*, 9(9).
- Visser, R. J., McDonagh, K. D., Meller, R. D., and McDonald, T. P. (2004). Harvesting system simulation using a systems dynamic model. *Southern Journal of Applied Forestry*, 28(2):91–99.
- Weintraub, A. and Bare, B. B. (1996). New issues in forest land management from an operations research perspective. *Interfaces*, 26(5):9–25.
- Weintraub, A. and Cholak, A. (1991). A hierarchical approach to forest planning. *Forest Science*, 37(2):439–460.